

S^n , ~~S^1~~ , RP^n , $\langle \dots \rangle$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$

$M \cup N$ $\wedge \vee$ $\langle X, S \rangle / \langle e^{2\pi i}, e^{2\pi i} \rangle$
 π_0, π_1^2 $\wedge \vee$ $\langle X, S \rangle / \langle e^{2\pi i}, e^{2\pi i} \rangle$

$\mathbb{Z} \times \mathbb{Z}$ $\langle a, b \mid aba = b \rangle$ free product.
 $ababab \dots$

$aba^2b^2 = e$ $\langle a, b \mid aba = ba \rangle$ $\mathbb{Z} \times \mathbb{Z}$
 $ab = ba \rightarrow$ group $\mathbb{Z} \times \mathbb{Z}$

$S^1 \vee S^2$ $\pi_1 = \mathbb{Z} \times \mathbb{Z}$ $\pi_2 = \mathbb{Z}$
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Theorem $M: n$ -dim $\pi_1(M) \cong \pi_1(M \cup \dots)$
 with $(n \geq 2)$ $\pi_1(RP^n) \cong \mathbb{Z}_2$ $\pi_1(S^n) = 0$

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